Implicit Differentiation and Inverse Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: First Derivative by Implicit Differentiation**

The graph of a differentiable function $y = f(x)$ is shown below. The tangent line to the graph at $x = 2$ has the equation $y = -x + k$. Find the value of k.

Final Answer: 5. The graph shows that the function $f(x)$ passes through the point $(2, 3)$. Therefore, $f(2) = 3$. The tangent line to the graph at $x = 2$ also passes through the point $(2, f(2))$, which is $(2, 3)$. The equation of the tangent line is given as $y = -x + k$. Substituting the point $(2, 3)$ into this equation, we get $3 = -2 + k$. Solving for k, we find $k = 3 + 2 = 5$.

2 **[Foundational] • Target: Second Derivatives Implicitly**

Consider the function $y = g(x)$ and its inverse $x = h(y)$. The tangent to the graph of $g(x)$ at $x = a$ has a gradient of $\frac{1}{3}$. If $g(a) = 5$, what is the gradient of the tangent to the graph of $h(y)$ at $y = 5$? Show your complete algebraic work.

Final Answer: 3. We are given that $y = g(x)$ and $x = h(y)$ are inverse functions. The gradient of the tangent to the graph of $g(x)$ at $x = a$ is given by $g'(a) = \frac{dy}{dx}|_{x=a}$. We are told that $g'(a) = \frac{1}{3}$. We are also given that $g(a) = 5$. Since $x = h(y)$ is the inverse function, it means that $h(g(x)) = x$ and $g(h(y)) = y$. The derivative of the inverse function $h(y)$ with respect to y is given by the formula $h'(y) = \frac{1}{g'(x)}$, where $y = g(x)$. We want to find the gradient of the tangent to the graph of $h(y)$ at $y = 5$. Since $g(a) = 5$, we have $y = 5$ corresponding to $x = a$. Therefore, we need to evaluate $h'(5)$. Using the inverse function theorem: $h'(5) = \frac{1}{g'(a)}$. Substituting the given value of $g'(a) = \frac{1}{3}$, we get $h'(5) = \frac{1}{\frac{1}{3}} = 3$. Thus, the gradient of the tangent to the graph of $h(y)$ at $y = 5$ is 3.

3 **[Foundational] • Target: Derivatives of Inverse Functions**

Find $\frac{d}{dx}[\sin((-2x-2)^3)]$.

Final Answer: $-6(-2x-2)^2\cos((-2x-2)^3)$. Chain rule (twice): $\cos((-2x-2)^3) \cdot 3(-2x-2)^2 \cdot -2 = -6(-2x-2)^2\cos((-2x-2)^3)$.

4 **[Foundational] • Target: Derivatives of Inverse Trigonometric Functions**

Find $\frac{d}{dx}[\sin((-2x-1)^3)]$.

Final Answer: $-6(-2x-1)^2\cos((-2x-1)^3)$. Chain rule (twice): $\cos((-2x-1)^3) \cdot 3(-2x-1)^2 \cdot -2 = -6(-2x-1)^2\cos((-2x-1)^3)$.

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5 [Foundational] • Target: First Derivative by Implicit Differentiation

Find $\frac{d}{dx}[(\sin(-2x) + -2)^3]$.

Final Answer: $-6(-\sin(2x) - 2)^2\cos(2x)$. Chain rule (twice): $3(\sin(-2x) + -2)^2 \cdot \cos(-2x) \cdot -2 = -6(-\sin(2x) - 2)^2\cos(2x)$.

6 [Foundational] • Target: Second Derivatives Implicitly

Find $\frac{d}{dx}[(\sin(-2x) + -1)^3]$.

Final Answer: $-6(-\sin(2x) - 1)^2\cos(2x)$. Chain rule (twice): $3(\sin(-2x) + -1)^2 \cdot \cos(-2x) \cdot -2 = -6(-\sin(2x) - 1)^2\cos(2x)$.

7 [Foundational] • Target: Derivatives of Inverse Functions

Find $\frac{d}{dx}[\sin((1-2x)^3)]$.

Final Answer: $-6(1-2x)^2\cos((1-2x)^3)$. Chain rule (twice): $\cos((1-2x)^3) \cdot 3(1-2x)^2 \cdot -2 = -6(1-2x)^2\cos((1-2x)^3)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Derivatives of Inverse Trigonometric Functions

Find $\frac{d}{dx}[\sin((-2x-2)^2)]$.

Final Answer: $(8x+8)\cos((-2x-2)^2)$. Chain rule (twice): $\cos((-2x-2)^2) \cdot 2(-2x-2)^1 \cdot -2 = (8x+8)\cos((-2x-2)^2)$.

9 [Intermediate] • Target: First Derivative by Implicit Differentiation

Find $\frac{d}{dx}[\sin((-2x-1)^2)]$.

Final Answer: $(8x+4)\cos((-2x-1)^2)$. Chain rule (twice): $\cos((-2x-1)^2) \cdot 2(-2x-1)^1 \cdot -2 = (8x+4)\cos((-2x-1)^2)$.

10 [Intermediate] • Target: Second Derivatives Implicitly

Find $\frac{d}{dx}[\sin((3x+1)^3)]$.

Final Answer: $9(3x+1)^2\cos((3x+1)^3)$. Chain rule (twice): $\cos((3x+1)^3) \cdot 3(3x+1)^2 \cdot 3 = 9(3x+1)^2\cos((3x+1)^3)$.

11 [Intermediate] • Target: Derivatives of Inverse Functions

Find $\frac{d}{dx}[(\sin(-2x) + 2)^3]$.

Final Answer: $-6(2 - \sin(2x))^2\cos(2x)$. Chain rule (twice): $3(\sin(-2x) + 2)^2 \cdot \cos(-2x) \cdot -2 = -6(2 - \sin(2x))^2\cos(2x)$.

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12 [Intermediate] • Target: Derivatives of Inverse Trigonometric Functions

Find $\frac{d}{dx}[\sin((3x-1)^3)]$.

Final Answer: $9(3x-1)^2\cos((3x-1)^3)$. Chain rule (twice): $\cos((3x-1)^3)\cdot 3(3x-1)^2\cdot 3 = 9(3x-1)^2\cos((3x-1)^3)$.

13 [Intermediate] • Target: First Derivative by Implicit Differentiation

Find $\frac{d}{dx}[\sin((2x+2)^3)]$.

Final Answer: $6(2x+2)^2\cos((2x+2)^3)$. Chain rule (twice): $\cos((2x+2)^3)\cdot 3(2x+2)^2\cdot 2 = 6(2x+2)^2\cos((2x+2)^3)$.

14 [Intermediate] • Target: Second Derivatives Implicitly

Find $\frac{d}{dx}[\sin((3x-2)^3)]$.

Final Answer: $9(3x-2)^2\cos((3x-2)^3)$. Chain rule (twice): $\cos((3x-2)^3)\cdot 3(3x-2)^2\cdot 3 = 9(3x-2)^2\cos((3x-2)^3)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Derivatives of Inverse Functions • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

Find $\frac{d}{dx}[\sin((2x+1)^3)]$.

Final Answer: $6(2x+1)^2\cos((2x+1)^3)$. Chain rule (twice): $\cos((2x+1)^3)\cdot 3(2x+1)^2\cdot 2 = 6(2x+1)^2\cos((2x+1)^3)$.

16 [Challenging] • Target: Derivatives of Inverse Trigonometric Functions • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

Find $\frac{d}{dx}[\sin((2x-1)^3)]$.

Final Answer: $6(2x-1)^2\cos((2x-1)^3)$. Chain rule (twice): $\cos((2x-1)^3)\cdot 3(2x-1)^2\cdot 2 = 6(2x-1)^2\cos((2x-1)^3)$.

17 [Challenging] • Target: First Derivative by Implicit Differentiation • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Find $\frac{d}{dx}[\sin((2x-2)^3)]$.

Final Answer: $6(2x-2)^2\cos((2x-2)^3)$. Chain rule (twice): $\cos((2x-2)^3)\cdot 3(2x-2)^2\cdot 2 = 6(2x-2)^2\cos((2x-2)^3)$.

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18 [Challenging] • Target: Second Derivatives Implicitly • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Find $\frac{d}{dx}[(\sin(-2x)+1)^3]$.

Final Answer: $-6(1-\sin(2x))^2\cos(2x)$. Chain rule (twice):

$3(\sin(-2x)+1)^2 \cdot \cos(-2x) \cdot -2 = -6(1-\sin(2x))^2\cos(2x)$.

19 [Challenging] • Target: Derivatives of Inverse Functions • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Find $\frac{d}{dx}[(\sin(-2x)+-1)^2]$.

Final Answer: $-4(-\sin(2x)-1)\cos(2x)$. Chain rule (twice):

$2(\sin(-2x)+-1)^1 \cdot \cos(-2x) \cdot -2 = -4(-\sin(2x)-1)\cos(2x)$.

20 [Challenging] • Target: Derivatives of Inverse Trigonometric Functions • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Find $\frac{d}{dx}[(\sin(-2x)+-2)^2]$.

Final Answer: $-4(-\sin(2x)-2)\cos(2x)$. Chain rule (twice):

$2(\sin(-2x)+-2)^1 \cdot \cos(-2x) \cdot -2 = -4(-\sin(2x)-2)\cos(2x)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: First Derivative by Implicit Differentiation

Find $\frac{dy}{dx}$ for the circle $x^2+y^2=25$.

a. $-\frac{x}{y}$ ✓

b. $\frac{x}{y}$

c. $-\frac{y}{x}$

d. $\frac{y}{x}$

Differentiate implicitly: $2x+2y\frac{dy}{dx}=0 \Rightarrow \frac{dy}{dx}=-\frac{x}{y}$.

22 [Foundational] • Target: First Derivative by Implicit Differentiation

Find $\frac{dy}{dx}$ for the curve $x^3+y^3=6xy$.

a. $\frac{2y-x^2}{y^2-2x}$ ✓

b. $\frac{x^2-2y}{2x-y^2}$

c. $\frac{2x-y^2}{x^2-2y}$

d. $\frac{x^2+y^2}{6}$

$3x^2+3y^2y'=6(y+xy') \Rightarrow 3y^2y'-6xy'=6y-3x^2 \Rightarrow y'(y^2-2x)=2y-x^2 \Rightarrow y'=\frac{2y-x^2}{y^2-2x}$.

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23 [Foundational] • Target: First Derivative by Implicit Differentiation

Find the slope of the tangent line to $x^2y + y^2 = 5$ at the point $(2, 1)$.

a. $-\frac{2}{3}$ ✓

b. $-\frac{4}{5}$

c. $-\frac{4}{6}$

d. -1

$2xy + x^2y' + 2yy' = 0$. At $(2, 1)$: $2(2)(1) + 4y' + 2(1)y' = 0 \Rightarrow 4 + 6y' = 0 \Rightarrow y' = -\frac{4}{6} = -\frac{2}{3}$.

24 [Foundational] • Target: Second Derivatives Implicitly

For $x^2 + y^2 = 16$, what is $\frac{d^2y}{dx^2}$ expressed solely in terms of y?

a. $-\frac{16}{y^3}$ ✓

b. $\frac{16}{y^3}$

c. $-\frac{1}{y^2}$

d. $-\frac{4}{y^3}$

$y' = -\frac{x}{y}$. Differentiate: $y'' = -\frac{1(y) - xy'}{y^2} = -\frac{y - x(-x/y)}{y^2} = -\frac{y^2 + x^2}{y^3}$. Since $x^2 + y^2 = 16$, $y'' = -\frac{16}{y^3}$.

25 [Foundational] • Target: Second Derivatives Implicitly

At the point $(0, 2)$ on the ellipse $4x^2 + y^2 = 4$, what is $\frac{d^2y}{dx^2}$?

a. -2 ✓

b. 2

c. -4

d. 0

$8x + 2yy' = 0 \Rightarrow y' = -\frac{4x}{y}$. At $(0, 2)$, $y' = 0$. $y'' = -\frac{4y - 4xy'}{y^2} = -\frac{4(2) - 0}{4} = -\frac{8}{4} = -2$.

26 [Foundational] • Target: Second Derivatives Implicitly

What is the sign of the concavity at $(3, 4)$ on the circle $x^2 + y^2 = 25$?

a. **Negative (concave down)** ✓

b. Positive (concave up)

c. Zero (inflection point)

d. Undefined

$y'' = -\frac{25}{y^3}$. At $(3, 4)$, $y'' = -\frac{25}{64} < 0$, which is concave down.

27 [Foundational] • Target: Derivatives of Inverse Functions

If g is the inverse of f, which formula gives $g'(a)$ where $f(b) = a$?

a. $g'(a) = \frac{1}{f'(b)}$ ✓

b. $g'(a) = f'(b)$

c. $g'(a) = \frac{1}{f'(a)}$

d. $g'(a) = -\frac{1}{f'(b)}$

By the Inverse Function Theorem, $g'(a) = \frac{1}{f'(f^{-1}(a))} = \frac{1}{f'(b)}$.

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28 [Foundational] • Target: Derivatives of Inverse Functions

Let $f(x) = x^3 + 2x - 1$. If g is the inverse of f, find $g'(2)$.

a. $\frac{1}{5}$ ✓

b. 5

c. $\frac{1}{14}$

d. 14

Find x where $f(x) = 2$: $x^3 + 2x - 1 = 2 \Rightarrow x^3 + 2x - 3 = 0 \Rightarrow x = 1$. Then $f'(x) = 3x^2 + 2 \Rightarrow f'(1) = 5$. Thus $g'(2) = \frac{1}{f'(1)} = \frac{1}{5}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Derivatives of Inverse Functions

Let $f(x) = x^5 + x + 1$. What is the derivative of f^{-1} at $x = 3$?

a. $\frac{1}{6}$ ✓

b. 6

c. $\frac{1}{80}$

d. $\frac{1}{5}$

Notice $f(1) = 1 + 1 + 1 = 3$. $f'(x) = 5x^4 + 1 \Rightarrow f'(1) = 6$. Therefore, $(f^{-1})'(3) = \frac{1}{6}$.

30 [Intermediate] • Target: Derivatives of Inverse Trigonometric Functions

What is $\frac{d}{dx}[\arcsin(3x)]$?

a. $\frac{3}{\sqrt{1-9x^2}}$ ✓

b. $\frac{1}{\sqrt{1-9x^2}}$

c. $\frac{3}{1+9x^2}$

d. $\frac{3}{\sqrt{1-3x^2}}$

Formula: $\frac{d}{dx}[\arcsin u] = \frac{u'}{\sqrt{1-u^2}}$. Here $u = 3x$, $u' = 3$, giving $\frac{3}{\sqrt{1-9x^2}}$.

31 [Intermediate] • Target: Derivatives of Inverse Trigonometric Functions

Find $\frac{d}{dx}[\arctan(x^2)]$.

a. $\frac{2x}{1+x^4}$ ✓

b. $\frac{1}{1+x^4}$

c. $\frac{2x}{1+x^2}$

d. $\frac{2x}{\sqrt{1-x^4}}$

Formula: $\frac{d}{dx}[\arctan u] = \frac{u'}{1+u^2}$. Here $u = x^2$, $u' = 2x$, giving $\frac{2x}{1+x^4}$.

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32 [Intermediate] • Target: Derivatives of Inverse Trigonometric Functions

What is the slope of the tangent line to $y = \arctan(x)$ at $x = 1$?

a. $\frac{1}{2}$ ✓

b. 1

c. $\frac{\pi}{4}$

d. $\frac{1}{\sqrt{2}}$

$y' = \frac{1}{1+x^2}$. At $x = 1$: $y'(1) = \frac{1}{1+1^2} = \frac{1}{2}$.

33 [Intermediate] • Target: First Derivative by Implicit Differentiation

Find $\frac{d}{dx}[(\sin(-2x) + 2)^2]$.

a. $4 - 2\sin(2x)$

b. $-4(2 - \sin(2x))\cos(2x)$ ✓

c. $4\sin(2x) - 8$

d. $(4 - 2\sin(2x))\cos(2x)$

Chain rule (twice): $2(\sin(-2x) + 2)^1 \cdot \cos(-2x) \cdot -2 = -4(2 - \sin(2x))\cos(2x)$.

34 [Intermediate] • Target: Second Derivatives Implicitly

Find $\frac{d}{dx}[\sin((3x - 2)^2)]$.

a. $(18x - 12)\sin((3x - 2)^2)$

b. $(18x - 12)\cos((3x - 2)^2)$ ✓

c. $(6x - 4)\cos((3x - 2)^2)$

d. $\cos((3x - 2)^2)$

Chain rule (twice): $\cos((3x - 2)^2) \cdot 2(3x - 2)^1 \cdot 3 = (18x - 12)\cos((3x - 2)^2)$.

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35 **[Intermediate]** • **Target: Derivatives of Inverse Functions**

Find $\frac{d}{dx}[(\sin(-2x)+1)^2]$.

a. $4\sin(2x)-4$

b. $-4(1-\sin(2x))\cos(2x)$ ✓

c. $2-2\sin(2x)$

d. $(2-2\sin(2x))\cos(2x)$

Chain rule (twice): $2(\sin(-2x)+1)^1\cdot\cos(-2x)\cdot -2=-4(1-\sin(2x))\cos(2x)$.

36 **[Intermediate]** • **Target: Derivatives of Inverse Trigonometric Functions**

Find $\frac{d}{dx}[\sin((3x+2)^2)]$.

a. $\cos((3x+2)^2)$

b. $(6x+4)\cos((3x+2)^2)$

c. $(18x+12)\sin((3x+2)^2)$

d. $(18x+12)\cos((3x+2)^2)$ ✓

Chain rule (twice): $\cos((3x+2)^2)\cdot 2(3x+2)^1\cdot 3=(18x+12)\cos((3x+2)^2)$.

37 **[Intermediate]** • **Target: First Derivative by Implicit Differentiation**

Find $\frac{d}{dx}[\sin((1-2x)^2)]$.

a. $(8x-4)\sin((1-2x)^2)$

b. $(2-4x)\cos((1-2x)^2)$

c. $\cos((1-2x)^2)$

d. $(8x-4)\cos((1-2x)^2)$ ✓

Chain rule (twice): $\cos((1-2x)^2)\cdot 2(1-2x)^1\cdot -2=(8x-4)\cos((1-2x)^2)$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Second Derivatives Implicitly • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Find $\frac{d}{dx}[\sin((3x+1)^2)]$.

a. $\cos((3x+1)^2)$

b. $(6x+2)\cos((3x+1)^2)$

c. $(18x+6)\cos((3x+1)^2)$ ✓

d. $(18x+6)\sin((3x+1)^2)$

Chain rule (twice): $\cos((3x+1)^2)\cdot 2(3x+1)^1\cdot 3 = (18x+6)\cos((3x+1)^2)$.

39 [Challenging] • Target: Derivatives of Inverse Functions • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Find $\frac{d}{dx}[\sin((3x-1)^2)]$.

a. $(18x-6)\sin((3x-1)^2)$

b. $(6x-2)\cos((3x-1)^2)$

c. $(18x-6)\cos((3x-1)^2)$ ✓

d. $\cos((3x-1)^2)$

Chain rule (twice): $\cos((3x-1)^2)\cdot 2(3x-1)^1\cdot 3 = (18x-6)\cos((3x-1)^2)$.

40 [Challenging] • Target: Derivatives of Inverse Trigonometric Functions • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Find $\frac{d}{dx}[\sin((2-2x)^2)]$.

a. $(4-4x)\cos((2-2x)^2)$

b. $(8x-8)\sin((2-2x)^2)$

c. $\cos((2-2x)^2)$

d. $(8x-8)\cos((2-2x)^2)$ ✓

Chain rule (twice): $\cos((2-2x)^2)\cdot 2(2-2x)^1\cdot -2 = (8x-8)\cos((2-2x)^2)$.

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41 [Challenging] • Target: First Derivative by Implicit Differentiation • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Find $\frac{d}{dx}[(\sin(3x) + \ -1)^3]$.

a. $3(\sin(3x) - 1)^2$

b. $9(\sin(3x) - 1)^2\cos(3x)$ ✓

c. $9(\sin(3x) - 1)^2$

d. $3(\sin(3x) - 1)^2\cos(3x)$

Chain rule (twice): $3(\sin(3x) + \ -1)^2 \cdot \cos(3x) \cdot 3 = 9(\sin(3x) - 1)^2\cos(3x)$.

42 [Challenging] • Target: Second Derivatives Implicitly • Cross-Lesson: 8.01 Area Between Curves

Find $\frac{d}{dx}[(\sin(2x) + \ -1)^3]$.

a. $3(\sin(2x) - 1)^2$

b. $6(\sin(2x) - 1)^2\cos(2x)$ ✓

c. $6(\sin(2x) - 1)^2$

d. $3(\sin(2x) - 1)^2\cos(2x)$

Chain rule (twice): $3(\sin(2x) + \ -1)^2 \cdot \cos(2x) \cdot 2 = 6(\sin(2x) - 1)^2\cos(2x)$.

43 [Challenging] • Target: Derivatives of Inverse Functions • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Find $\frac{d}{dx}[\sin((2x - 1)^2)]$.

a. $\cos((2x - 1)^2)$

b. $(8x - 4)\sin((2x - 1)^2)$

c. $(8x - 4)\cos((2x - 1)^2)$ ✓

d. $(4x - 2)\cos((2x - 1)^2)$

Chain rule (twice): $\cos((2x - 1)^2) \cdot 2(2x - 1)^1 \cdot 2 = (8x - 4)\cos((2x - 1)^2)$.

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44 [Challenging] • Target: Derivatives of Inverse Trigonometric Functions • Cross-Lesson: 1.01 Limits Graphically and Numerically

Find $\frac{d}{dx}[\sin((2x-2)^2)]$.

a. $(8x-8)\sin((2x-2)^2)$

b. $(8x-8)\cos((2x-2)^2)$ ✓

c. $(4x-4)\cos((2x-2)^2)$

d. $\cos((2x-2)^2)$

Chain rule (twice): $\cos((2x-2)^2)\cdot 2(2x-2)^1\cdot 2=(8x-8)\cos((2x-2)^2)$.

45 [Challenging] • Target: First Derivative by Implicit Differentiation • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Find $\frac{d}{dx}[(\sin(3x)+\ -2)^3]$.

a. $3(\sin(3x)-2)^2\cos(3x)$

b. $9(\sin(3x)-2)^2$

c. $3(\sin(3x)-2)^2$

d. $9(\sin(3x)-2)^2\cos(3x)$ ✓

Chain rule (twice): $3(\sin(3x)+\ -2)^2\cdot\cos(3x)\cdot 3=9(\sin(3x)-2)^2\cos(3x)$.

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